PHILIPPINE STARTUP CHALLENGE XI

WeCare

RoadGuard

CONCEPT NOTE

I. SUMMARY

RoadGuard is a mobile application designed to enhance road safety by minimizing distractions caused by texting while driving. The app automatically disables manual keyboard input when driving is detected, allowing only voice-to-text, voice messaging, and calls. This ensures drivers keep their eyes on the road and hands on the wheel while maintaining communication.

For passenger safety, RoadGuard includes an SOS feature, automatic impact detection, and live tracking with real-time notifications sent to parents or loved ones.

Expected outcomes include reduced text-related distractions, improved driving focus, fewer traffic violations, and greater peace of mind for families through enhanced passenger safety monitoring.

II. BACKGROUND OF THE PROBLEM

Cell phone use while driving remains a major safety concern, as many drivers continue to text or make calls on the road, increasing the risk of accidents. Al-Ajlouny and Alzboon (2023) found that frequent mobile phone users are significantly more likely to engage in risky behaviors such as calling, texting, or browsing while driving, and those who always use their phones while driving commit more traffic violations compared to those who rarely or never do. Additionally, road safety is increasingly compromised by distracted driving, which significantly elevates crash risk. Distractions can be visual (eyes off the road), manual (hands off the wheel), or cognitive and acoustic (attention diverted from driving) according to Voinea, Boboc, Buzdugan, Antonya, and Yannis (2023). Current solutions fail to fully restrict manual input, leaving drivers vulnerable to distraction.This app enforces voice-only interaction to ensure safer communication while driving.

III. PROPOSED STARTUP SOLUTION

A mobile application designed to enhance road safety by minimizing distractions caused by texting while driving. This app would automatically disable manual keyboard input when driving is detected, allowing only voice-to-text, voice messaging, and calls. This ensures drivers keep their eyes on the road and hands on the wheel while maintaining communication. This solution focuses on SDG 3, specifically Target 3.6, to halve the number of global deaths and injuries from road traffic accidents.

IV. OBJECTIVES

- Reduce text-related distractions among users by 80% within 6 months of app adoption.
- Implement driving detection with 95% reliability at launch.
- Achieve 95% accuracy in voice-to-text and ensure stable call features within 12 months of release.
- Reach 500 pilot users within the first year.

V. TARGET MARKET/BENEFICIARIES

Rideshairing Companies (Move It, Grab, etc.)

VI. VALUE PROPOSITION

- Safer driving habits and reduced accident risks.
- Lower incidence of traffic violations among users.
- Hands-free call control system and voice-based call handling.
- Driving detection module that automatically disables the keyboard when driving but includes an emergency override.

VII. BUSINESS MODEL

B2B Enterprise Licensing -

VIII. MARKET ANALYSIS

Assessment of the target market, including size, demand, trends, and competition.

IX. OPERATIONS PLAN

Overview of how the startup will implement and deliver its product or service.

X. FINANCIAL REQUIREMENT

Item / CategoryItem / Category	Cost (PHP)
Development	₱300,000

Testing	₱80,000
Marketing	₱120,000
Contingency (10%)	₱50,000
Total:	₱550,000